

20 Data Cleaning Scenarios for Data Analysts

Practical real-world scenarios every data analyst faces — with Excel solutions.

Scenario 1: Extra Spaces in Data

Task: Customer names and IDs have extra leading/trailing spaces.

Solution: Use =TRIM(A2) or Power Query → Transform → Format → Trim.

Scenario 2: Inconsistent Capitalization

Task: Names are in random upper/lower cases.

Solution: Use =PROPER(A2); Power Query → Format → Capitalize Each Word.

Scenario 3: Duplicate Records

Task: Multiple identical rows appear.

Solution: Data → Remove Duplicates or Power Query → Remove Duplicates.

Scenario 4: Blank Cells

Task: Missing emails or sales values.

Solution: Filter blanks or Power Query → Remove Rows → Remove Blank Rows.

Scenario 5: Mixed Date Formats

Task: Dates in text, numbers, or various formats.

Solution: Text to Columns → Date or Power Query → Change Type → Date.

Scenario 6: Non-Printable Characters

Task: Imported text has invisible characters.

Solution: Use =CLEAN(A2) or Power Query → Transform → Clean.

Scenario 7: Merged Cells

Task: Merged headers/values prevent sorting.

Solution: Unmerge cells and Fill Down in Power Query.

Scenario 8: Misaligned Headers

Task: Headers start from row 2.

Solution: Delete extra rows; Power Query → Use First Row as Headers.

Scenario 9: Trailing Punctuation

Task: Names end with commas/periods.

Solution: Use =SUBSTITUTE(A2,",","") or Power Query → Replace Values.

Scenario 10: Multiple Tables in One Sheet

Task: Sheet contains multiple unrelated tables.

Solution: Load each table separately in Power Query; append or merge as needed.

Scenario 11: Numbers Stored as Text

Task: Numeric values are left-aligned as text.

Solution: Use =VALUE(A2) or multiply by 1; Power Query → Detect Data Type.

Scenario 12: Inconsistent Category Names

Task: 'NY', 'New York', 'N.Y.' appear in the same column.

Solution: Use VLOOKUP mapping table or Power Query → Replace Values.

Scenario 13: Extra Line Breaks

Task: Cells have unwanted line breaks.

Solution: Use =SUBSTITUTE(A2,CHAR(10)," ") or Power Query → Replace Values.

Scenario 14: Currency Symbols Mixed

Task: Some cells have \$ or ■ mixed with numbers.

Solution: Remove symbols using SUBSTITUTE or Power Query → Replace Values.

Scenario 15: Negative Numbers as Text

Task: Values like (123) need to be converted to -123.

Solution: Use Excel formulas or Power Query → Transform → Standardize.

Scenario 16: Leading Zeros Lost

Task: IDs like 00123 converted to 123.

Solution: Use =TEXT(A2,"00000") or Power Query → Transform → Format → Text.

Scenario 17: Inconsistent Boolean Values

Task: Yes/No vs True/False vs 1/0 in same column.

Solution: Standardize with IF formulas or Power Query → Replace Values.

Scenario 18: Hyperlinks in Cells

Task: Imported cells have embedded hyperlinks.

Solution: Right-click → Remove Hyperlink or Power Query → Transform → Extract Text.

Scenario 19: Special Characters in Names

Task: Names include #, %, &, etc.

Solution: Use =SUBSTITUTE multiple times or Power Query → Replace Values.

Scenario 20: Multiple Word Separators

Task: Columns separated by -, _, or / inconsistently.

Solution: Use Text to Columns or Power Query → Split Column → By Delimiter.